Outline

Probability Theory Refreshers

1. **Uniform Random Variable**

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- A continuous random variable X is said to be **uniformly distributed** on the interval $[a, b]$ if its Probability Density Function (PDF) is defined as:

$$X \sim \mathcal{U}(a, b) \Leftrightarrow f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- Its variance is $\mathrm{Var}(X) = \frac{(b-a)^2}{12}$

2. Expected Value of a Function of an R.V. (LOTUS)

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2. Expected Value of a Function of an R.V. (LOTUS)

- Let X be a continuous random variable with PDF $f_X(x)$. The expected value of a real-valued function $g(X)$ is given by:

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

- This theorem (Law of the Unconscious Statistician) allows us to calculate the mean of $g(X)$ without first finding its own PDF.

Uniform Quantization: RD curve

Let us apply these results to find the distortion of a uniform RV quantized with UQ.
Hypotheses : $X \sim \mathcal{U}\left(-\frac{A}{2}, \frac{A}{2}\right)$ and $Q(x)$ is a uniform quantizer with L levels.
Goal: Find $\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[(X - \widehat{X})^2\right]$ as a function of A, L

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We observe that $\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[g(X)\right]$, where $\forall u \in \mathbb{R}, g(u) = (u - Q(u))^2$.

$$\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[(X-\widehat{X})^2\right] = \int_{-\infty}^{+\infty} p_X(u)[u-Q(u)]^2 du$$

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$$\sigma_{\mathrm{Q}}^2 = \mathrm{E}\left[(X - \widehat{X})^2\right] = \int_{-\infty}^{+\infty} p_X(u)[u - Q(u)]^2 du$$

$$\ldots = \sum_{i=1}^{L} \int_{\Theta^i} \frac{1}{A}[u - \widehat{x}^i)]^2 du = \frac{1}{A} \sum_{i=1}^{L} \int_{\widehat{x}^i - \Delta/2}^{\widehat{x}^i + \Delta/2} [u - \widehat{x}^i)]^2 du$$

$$= \frac{1}{A} \sum_{i=1}^{L} \int_{-\Delta/2}^{\Delta/2} t^2 dt = \frac{1}{A} L \frac{\Delta^3}{12} = \frac{\Delta^2}{12} = \frac{1}{12} \frac{A^2}{L^2}$$

Uniform Quantization: RD curve

We can use the previous result to determine the **rate-distortion** curve of the UQ, i.e. the mathematical relationship between the rate R and the distortion $D = \sigma_Q^2$.

$$D = \frac{\Delta^2}{12} \qquad = \frac{A^2}{12L^2} \qquad = \frac{A^2/12}{\cdot 2^{2R}} \qquad = \sigma_X^2 2^{-2R}$$

$$\mathrm{SNR} = 10 \log_{10} \frac{\mathrm{E}\left\{X^2\right\}}{D} \qquad = 10 \log_{10} \frac{\sigma_X^2}{\sigma_X^2 2^{-2R}}$$

$$= 10 \log_{10} 2^{2R} \approx 6R$$

High Resolution (HR) Uniform Quantization

- Hypothesis : $L \to +\infty$, X generic RV
- In HR, in any given Θ^i we approximate p_X as a constant (it is constant in the region, but the value can be different in another region)
- Therefore, the quantization noise in Θ^i is $\mathcal{U}\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$
- From the total probability law, $E \sim \mathcal{U}\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$
- Thus:

$$D = \frac{\Delta^2}{12} = \frac{A^2}{12} 2^{-2R}$$

High Resolution (HR) Uniform Quantization

$$\mathrm{SNR} = 10\log_{10}\frac{\mathrm{E}\left\{X^2\right\}}{D} \qquad = 10\log_{10}\frac{\sigma_X^2}{A^2/12}2^{2R} \qquad \approx 6R - 10\log_{10}\frac{\gamma^2}{3}$$

where $\gamma^2 = \frac{X_{\mathrm{max}}^2}{\sigma_X^2} = \frac{A^2/4}{\sigma_x^2}$ is the *load factor*, i.e. the ratio between the peak power and the average power.

$$D = \frac{A^2}{3}2^{-2R} = \frac{\gamma^2}{3}\sigma_X^2 2^{-2R} = K_X\sigma_X^2 2^{-2R}$$

Outline

Optimal quantization

For a given probability density function $p_X(x)$, we want to find the quantizer minimizing distortion for a given rate
Problem equivalent to determine thresholds t^i and levels $\widehat{x}^i$.
Solutions :

- Analytic Solution in *high resolution*: $D = c_X \sigma_x^2 2^{-2R}$
- If the high resolution hypothesis is not met, we can find a local optimum with the Lloyd-Max algorithm

Optimal quantization in high resolution

- The optimal quantizer is the one that, for a given PDF p_X of the input samples, provides the least distortion $D = \mathrm{E}\left[(X - Q(X))^2\right]$ for a given rate R
- It is possible to find the *rate-distortion* curve for the optimal quantizer in the **high-resolution hypothesis**:
  - $L \to \infty$
  - $\max_i \Delta_i \to 0$
  - $\forall i, \forall u \in \Theta^i, p_X(u) \approx P_i$
- Then it can be shown that the optimal quantizer has the following RD curve:

$$\sigma_{\mathrm{Q}}^2 = c_X \sigma_X^2 2^{-2R} \qquad \text{with} \quad c_X = \frac{1}{12}\left[\int_{\mathbb{R}} p_U^{1/3}(t)dt\right]^3 \qquad \text{and} \quad U = \frac{X}{\sigma_X}$$

Shape Factor

In the RD curve of the optimal qunatizer

$$\sigma_Q^2 = c_X \sigma_X^2 2^{-2R}$$

the term c_X is called *shape factor*, since it only depends on the PDF *shape*, and not on its variance

The shape of X is obviously the same as of U

Shape factors for common PDFs:

	Uniform	Gaussian
c_X	1	$\frac{\sqrt{3}}{2}\pi \approx 2.72$

Non-HR Optimal quantization

- High resolution hypothesis is not always met
- On the contrary, often quantization must work at low or very low rate
- **There is no analytical solution for low resolution optimal quantization**
- On the other hand, it is possible to find **necessary conditions** for optimal quantizers independently from the resolution
- These conditions allows to define an algorithm attaining a local optimum point (Lloyd-Max algorithm)

Optimal quantization

Lloyd-Max algorithm

1. Choose an initial dictionary $\mathcal{C}^{(0)} = \{\widehat{x}_0^i\}_{i=1,\dots L}$
2. Find the *best* thresholds for the given dictionary
3. Find the *best* dictionary for the given thresholds
4. Iterate 2-3 until a stop condition is met

Lloyd-Max algorithm: initial dictionary

- Uniform Initialization: $\widehat{x}^n = -A_0 + n\Delta \; \forall n = 1, \ldots, L$
- Random Initialization: code-words are initialized with the same distribution as data

Lloyd-Max algorithm: nearest neighbor rule

- Given $\{\widehat{x}^i\}_{i=1,\dots L}$, we have to define, for each input, which is the best level to use as its representative
- The best choice is obviously the nearest neighbor:

$$k = \arg\min_n |x - \widehat{x}^n| \Rightarrow Q(x) = \widehat{x}^k$$

- Choosing the next neighbor is equivalent to set the threshold at the midpoint between consecutive quantization levels:

$$t^i = \frac{\widehat{x}^i + \widehat{x}^{i+1}}{2}, \quad i \in \{1, \dots, L-1\}$$

Lloyd-Max algorithm: centroid condition

Given the thresholds $\{t^0, \dots t^L\}$, the best levels are those minimizing the distortion:

$$\sigma_Q^2 = \sum_{i=1}^{L} \int_{t^{i-1}}^{t^i} (u - \widehat{x}^i)^2 p_X(u) du$$

To find the optimal levels, we compute the gradient of the distortion and set it to 0.

$$\begin{aligned}
\frac{\partial \sigma_Q^2}{\partial \widehat{x}^i} &= \frac{\partial}{\partial \widehat{x}^i} \int_{t^{i-1}}^{t^i} (u - \widehat{x}^i)^2 p_X(u) du = \int_{t^{i-1}}^{t^i} \frac{\partial}{\partial \widehat{x}^i} (u - \widehat{x}^i)^2 p_X(u) du \\
&= \int_{t^{i-1}}^{t^i} 2(\widehat{x}^i - u) p_X(u) du = 2\widehat{x}^i \int_{t^{i-1}}^{t^i} p_X(u) du - 2 \int_{t^{i-1}}^{t^i} u p_X(u) du \\
\widehat{x}^i &= \frac{\int_{t^{i-1}}^{t^i} u p_X(u) du}{\int_{t^{i-1}}^{t^i} p_X(u) du} = \frac{\int_{\Theta_i} u p_X(u) du}{\int_{\Theta_i} p_X(u) du} = \mathrm{E}\left[X | X \in \Theta_i\right]
\end{aligned}$$

This is the so-called centroid condition: the best representative of a quantization region is its centroid, i.e. the conditional average of X given $X \in \Theta^i$

Lloyd-Max algorithm: stop condition

- After the k-iteration of NN and the centroid rules, we can compute the new distortion value $\sigma^2_{\mathrm{Q},(k)}$
- It can be compared with the previous distortion $\sigma^2_{\mathrm{Q},(k-1)}$: the optimization steps assure that the new distortion can only be less than or equal to the old one
- We compute the relative distortion reduction: if it is smaller than a threshold, we stop the algorithm:
$$\frac{\sigma^2_{\mathrm{Q},(k-1)} - \sigma^2_{\mathrm{Q},(k)}}{\sigma^2_{\mathrm{Q},(k-1)}} < \varepsilon$$
- Another stop condition is upon a given number of iterations:
$$k = K$$

Convergence of Lloyd-Max algorithm

- Convergence to global optimum is not assured
- Final result depends on **initialization**
- However, at each step distortion cannot increase

Lloyd-Max algorithm for data

- In real-life problems, we have not signal PDFs, but rather a bunch of data
- We can modify Lloyd-Max algorithm as follows:
  1. Let $\mathcal{X} = \{u_1, u_2, \dots u_M\}$ be the set of data to quantize (or the training set if data is not known in advance)
  2. Initialize (k=0) with any dictionary (e.g. uniform): $\mathcal{C}^{(k)} = \{\widehat{x}_0^i\}_{i=1,\dots L}$
  3. Nearest neighbor rule:
  $$W_k^i = \{u_m \in \mathcal{X} : \forall j \neq i \|u_m - \widehat{x}_k^i\| \leq \|u_m - \widehat{x}_k^j\|\}$$
  4. Centroid rule: $\widehat{x}_{k+1}^i = \frac{1}{|W_k^i|} \sum_{u_m \in W_k^i} u_m$
  5. Iterate to step 3 until convergence

Scalar quantization: summary

- $D(R)$, UQ of uniform RV: $D = \sigma_X^2 2^{-2R}$
- $D(R)$, UQ of non uniform RV in HR: $D = K_X \sigma_X^2 2^{-2R}$
- D(R), OQ in HR (any RV): $\sigma_{\mathrm{Q}}^2 = c_X \sigma_X^2 2^{-2R}$
- c_X is the shape factor
  - $c_X = 1$ for uniform RVs
  - $c_X = \frac{\sqrt{3}}{2}\pi$ for Gaussian RVs
- At any resolution, the LMA produces a non-uniform quantizer which is locally optimal